Loss reweighting moves a frozen representation's predictions but not its discrimination

Balanced MSE and inverse-density reweighting against the stabilizing-mutation deficit · 12,359 held-out mutations over 412 proteins · cluster-bootstrapped paired differences

Balanced MSE removes 19 % of the stabilizing bias and improves nothing else. The bias on stabilizing mutations falls from +0.58 to +0.47 kcal/mol, an interval that clearly excludes zero. Every ranking metric on that same class — ρ, AUC, top-K detection precision — is statistically indistinguishable from baseline, while overall Pearson r and MAE get significantly *worse*. Inverse-density reweighting is dominated: no significant gain anywhere, significant losses on ρ and MAE. **Neither loss is adopted.**

1. Motivation

A per-class error breakdown of this predictor found one deficit that survives normalisation by class effect size: it systematically calls stabilizing mutations destabilizing. On held-out data the bias on that class is +0.58 kcal/mol and its within-class Spearman ρ is only 0.26, against a class that makes up 4.3 % of the corpus (535 of 12,359 mutations). That is precisely the regime protein engineering cares about — the goal is to *find* the stabilizing mutation, not to rank the destabilizing ones.

Label-imbalance losses are the cheapest published lever for exactly this shape of problem: no new features, no architecture change, no retraining of the upstream representation. A recent constraint-aware stability predictor reports a substantial external-benchmark gain from loss-level changes alone, with Balanced MSE the term aimed at the rare tail. The question here is whether that lever works when the representation underneath is **frozen**.

2. Methods

Three losses, identical architecture, data, splits and seeds.

- **Plain MSE** — the project baseline.
- **Balanced MSE**, Monte-Carlo form — the batch is treated as a classification over which target each prediction belongs to, which divides out the training label density. The noise variance is learned in log space.
- **LDS** — inverse smoothed-density sample weighting over a 40-bin ΔΔG histogram.

Both reweighted losses require sample weights or a custom objective, which the project's default estimator does not support, so all three were reimplemented in torch at the same topology and with the same antisymmetry augmentation. Evaluation is 5-fold grouped cross-validation on wild-type identity, two seeds averaged.

Statistics. A cluster bootstrap over the 412 proteins, reported as the *paired* difference against MSE computed within each resample. The pairing matters more than the bootstrap: the absolute intervals for the three losses overlap heavily because they carry the between-protein variance that pairing cancels.

The stabilizing class is defined as ΔΔG < −0.5 kcal/mol throughout, and is scored on four separate axes — bias (are the magnitudes right?), within-class ρ and AUC (is the ordering right?), and top-K detection precision (would a screen find them?). The distinction between the first and the rest is the result.

3. Results

metric	Balanced MSE	LDS
bias, stabilizing class	**-0.113 [-0.144, -0.080] ***	+0.020 [-0.011, +0.052]
ρ, stabilizing class	+0.005 [-0.034, +0.044]	-0.008 [-0.057, +0.040]
AUC, stabilizing class	+0.003 [-0.004, +0.011]	-0.010 [-0.022, +0.002]
detection precision @30	+0.017 [-0.133, +0.167]	-0.005 [-0.167, +0.167]
ρ overall	+0.003 [-0.001, +0.008]	**-0.011 [-0.018, -0.005] ***
r overall	**-0.052 [-0.131, -0.001] ***	-0.042 [-0.116, +0.005]
MAE overall	**+0.088 [+0.076, +0.100] ***	**+0.023 [+0.015, +0.034] ***

Paired difference against MSE with 95 % cluster-bootstrap CI over 412 proteins, on held-out data. Bold and starred intervals exclude zero. For bias and MAE, negative is better; for every other row, positive is better.

3.1 The bias is fixed; the blindness is not

Balanced MSE is the only arm with a real gain, and it is confined to one row. The stabilizing-class bias moves **-0.113 [-0.144, -0.080] *** — from +0.58 to +0.47 kcal/mol. Every measure of whether the model can *tell which* mutations stabilize is unchanged: within-class ρ +0.005 [-0.034, +0.044], AUC +0.003 [-0.004, +0.011], detection precision +0.017 [-0.133, +0.167].

The cost is not confined. Overall Pearson r moves **-0.052 [-0.131, -0.001] *** and MAE **+0.088 [+0.076, +0.100] ***, both significantly in the wrong direction — the reweighting buys the tail's calibration by spending accuracy on the bulk.

LDS is dominated outright: **-0.011 [-0.018, -0.005] *** on overall ρ and **+0.023 [+0.015, +0.034] *** on MAE, both significant losses, with no significant gain on any tail metric.

3.2 What the figure shows

The mechanism is visible directly. Balanced MSE shifts stabilizing predictions downward as a group, which is what a bias correction looks like. It does not change *which* mutations are ranked most stabilizing — the top-K precision curves for the three losses lie on top of one another.

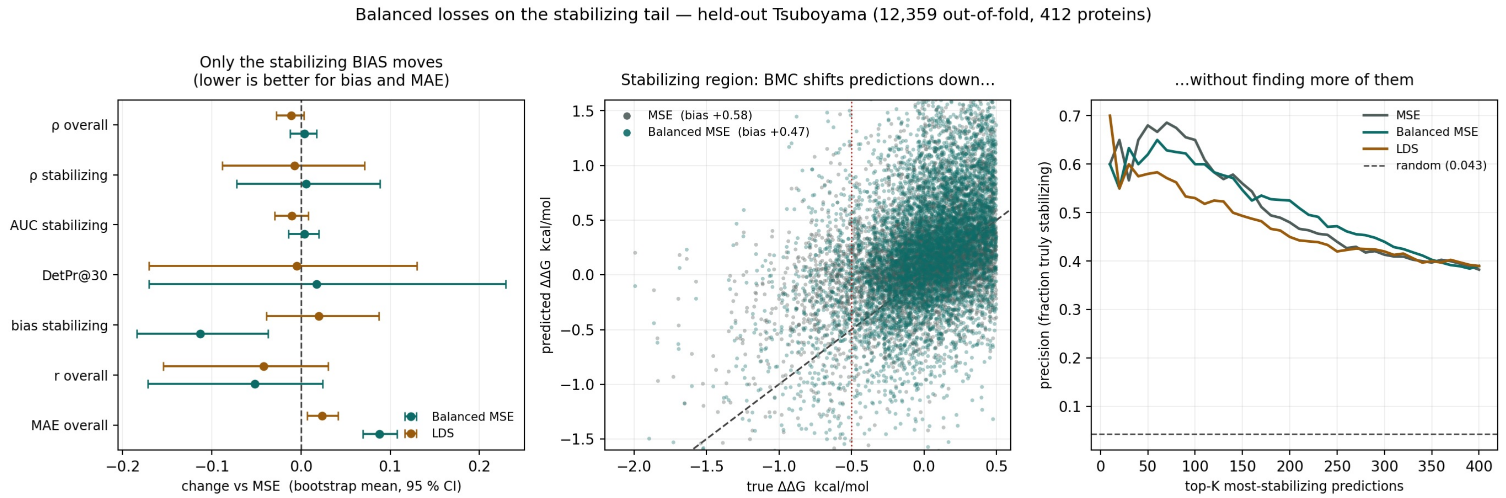

Figure 1. Left: paired differences against MSE with their intervals — only the stabilizing bias clears zero in the good direction, while r and MAE clear it in the bad one. Middle: the stabilizing region, showing the downward shift. Right: precision among the top-K predicted-most-stabilizing mutations; the three curves are superimposed, so the shift finds no additional stabilizing mutations.

3.3 The same losses on an external benchmark

Transferred without refitting to a blind external benchmark, no loss separates from any other: overall r is 0.387 / 0.368 / 0.383 for MSE / Balanced MSE / LDS, and the within-tail ρ collapses toward zero for all three (+0.081 / -0.001 / +0.056). That is consistent with the separate finding that the error on that benchmark is dominated by cross-dataset domain shift rather than by the tail — a loss term aimed at the tail cannot address it.

4. Interpretation

With a frozen representation, the loss can move predictions but cannot create discrimination the features do not carry. The bias is a property of the objective, and it is fixable by changing the objective. The inability to *identify* which mutations stabilize is a property of the representation, and it is not.

This reading also explains why the published precedent and this result differ without either being wrong. The constraint-aware predictor that gains from the same Balanced-MSE term **fine-tunes its backbone**. There, the representation can adapt to the reweighted objective — the loss changes what the features learn to encode. Here the trunk is frozen and the loss only redistributes predictions within a fixed feature space.

The practical consequence is a recommendation and a condition on it. Neither loss should be adopted as a default: the only real gain costs pooled r and MAE, and it does not improve the metric that matters for engineering, which is finding stabilizing mutations. Balanced MSE is worth revisiting *after* the representation is unfrozen, which is the setting where the published precedent actually applies.

5. Limitations

- The stabilizing class holds 535 mutations. Top-K detection precision in particular is noisy at this size — its interval, +0.017 [-0.133, +0.167], is wide enough to be uninformative in both directions, and no conclusion rests on it.
- Two seeds per configuration, averaged. Seed variance is not separated from the between-protein variance the bootstrap captures.
- Only two reweighting schemes were tested, at their published default settings. A negative result for these two is not a negative result for label-imbalance methods in general.
- The stabilizing threshold of −0.5 kcal/mol is a convention. It was not varied, so the findings are stated for that class definition rather than for a continuum of tail severities.

6. Conclusion

Asked whether reweighting the loss toward a rare, badly predicted tail can fix that tail when the representation feeding it is frozen, the answer separates cleanly into two halves. The systematic offset on the tail is an objective-level problem and reweighting removes a fifth of it. Which mutations belong in the tail is a representation-level problem, and reweighting does not touch it — at a measurable cost to the bulk. The deficit stays open, and the direction it points is the representation, not the loss.